

Project Design Phase

Solution Architecture

Project Name : FitBuddy

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

FitBuddy Solution Architecture & Data Flow:

FitBuddy solution architecture defines the interaction between the user interface, backend processing service, health prediction models, and cloud database storage.

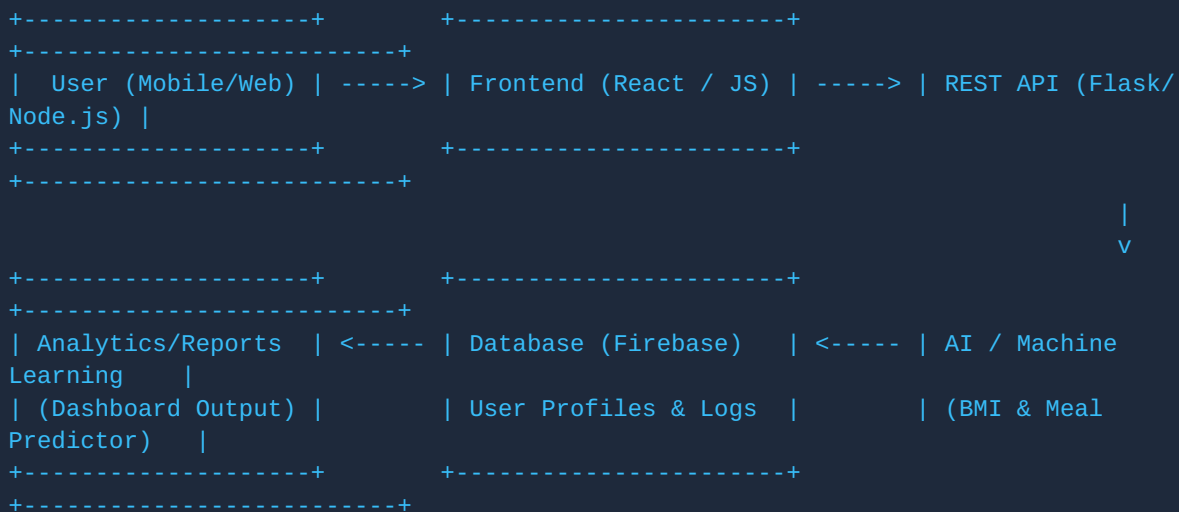


Figure 1: System Architecture and end-to-end data flow for FitBuddy application

Architecture Components & Technical Details:

Layer	Technology	Role in FitBuddy Architecture
Presentation Layer	HTML5, CSS3, JavaScript / React	Provides intuitive user interfaces for step tracking, workout logging, and health dashboards.
Application Layer	Python (Flask / FastAPI)	Handles user authentication, business logic, calorie math, and API communication.
Intelligence Layer	Scikit-Learn / ML Models	Processes user stats (height, weight, activity level) to recommend personalized diet and workouts.
Data Layer	Firebase / SQLite	Stores encrypted user credentials, daily step counts, calorie burn logs, and fitness milestones.